fertilizer.frt

Purpose

`fertilizer.frt` lists every fertilizer product the simulation can apply, with its mass fractions of mineral nitrogen, mineral phosphorus, organic nitrogen, organic phosphorus, and the share of mineral N that is in the NH3 form. Each record fills one `fertilizer_db` entry in the `fertdb` array. Fertilizer operations (in `chem_app.ops` and the management decision tables) reference entries here by name.

Source

- Reader: `src/fert_parm_read.f90`
- Type definition: `src/fertilizer_data_module.f90`, `type fertilizer_db`

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one record per fertilizer product. List-directed read into `fertdb(it)` (`type fertilizer_db`).

#	Field	Type	Units	Default	Description
1	fertnm	char(16)	none	“”	fertilizer name (key referenced by `chem_app.ops` and management decision tables)
2	fminn	real	kg min-N / kg fertilizer	0.0	fraction of fertilizer that is mineral N (NO3 + NH3)
3	fminp	real	kg min-P / kg fertilizer	0.0	fraction of fertilizer that is mineral P
4	forgn	real	kg org-N / kg fertilizer	0.0	fraction of fertilizer that is organic N
5	forgp	real	kg org-P / kg fertilizer	0.0	fraction of fertilizer that is organic P
6	fnh3n	real	kg NH3-N / kg min-N	0.0	fraction of mineral N content that is NH3

The five real fractions must sum to less than or equal to 1 for a single product. `fminn + fminp + forgn + forgp` is the total nutrient mass per unit fertilizer; the remainder is carrier mass.

Example

First lines of `refdata/Ames_sub1/fertilizer.frt`:

```
fertilizer.frt: written by SWAT+ editor v2.2.2 on 2023-10-17 14:43 for SWAT+ rev.60.5.4
name                      min_n         min_p         org_n         org_p         nh3_n          pathogens  c
elem_n                  1.00000       0.00000       0.00000       0.00000       0.00000               null  l
elem_p                  0.00000       1.00000       0.00000       0.00000       0.00000               null  l
anh_nh3                 0.82000       0.00000       0.00000       0.00000       1.00000               null  A
urea                    0.46000       0.00000       0.00000       0.00000       1.00000               null  U
46_00_00                0.46000       0.00000       0.00000       0.00000       0.00000               null  4
18_46_00                0.18000       0.20200       0.00000       0.00000       0.00000               null
dairy_fr                0.00700       0.00500       0.03100       0.00300       0.99000               null  l
```

Notes on the example:

- The editor writes two trailing columns (`pathogens` and `description`) that are not in `fertilizer_db`. List-directed reads stop once the type is filled, so both trailing tokens are discarded by the reader. The literal `null` in the `pathogens` column has no effect.
- For `urea`, `fminn = 0.46` and `fnh3n = 1.00`. That is, 46% of the product is mineral N, and 100% of that mineral N is in the NH3 form.
- For `18_46_00`, `fminn = 0.18` and `fminp = 0.202`. Note that the third value in the "46" position is the P (P2O5 converted to P), not P2O5 directly.

manure_db.frt

Named manure types. Each entry points to one organic-matter record in `manure_om.frt` and (optionally) to records for pesticides, pathogens, heavy metals, salts, and other constituents that may be carried with the manure. This is the lookup used by manure-application operations and by manure-allocation.md.

Source

- Reader: src/manure_db_read.f90
- Type: src/fertilizer_data_module.f90, `type manure_database` (variable `manure_db`)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one record per manure type. Two passes: count rows, then read.

#	Field	Type	Description
1	name	char(25)	Manure-type name. Referenced by manure-application operations and by `manure_allo.mnu`.
2	org_min	char(25)	Crosswalk into `manure_om.frt`. Resolved into `iorg_min`.
3	pests	char(25)	Pesticide profile name. Pointer field `ipests` is reserved (the crosswalk loop is commented out in the reader). Use `null` when not applicable.
4	paths	char(25)	Pathogen profile name. Pointer `ipaths`.
5	hmets	char(25)	Heavy-metal profile name. Pointer `imets`.
6	salts	char(25)	Salt-ion profile name. Pointer `isalts`.
7	constit	char(25)	Other-constituent profile name. Pointer `iconstit`.
8	descrip	char(80)	Free-text description.

After reading each row, the reader crosswalks `org_min` against `manure_om(:)%name` and stores the matching index in `manure_db(it)%iorg_min`. The other crosswalks are present in the type but not wired in this reader.

Example

`refdata/Osu_1hru/manure_db.frt`:

```
manure_db.frt: manure database
name          org_min             pests  paths  hmets  salts  constit  description
beef_salt     mw_beef_liq         null   null   null   null   null     midwest_beef_liquid_salt
```

One manure type, pointing at `mw_beef_liq` in `manure_om.frt`. No pesticide, pathogen, heavy-metal, salt, or constituent profile.

manure_om.frt

Organic-matter attributes for each manure type. Holds the water fraction and the C, mineral N, mineral P, organic N, organic P, and NH3 fractions used when a manure application is partitioned into the soil pools.

Source

- Reader: src/manure_orgmin_read.f90
- Type: src/fertilizer_data_module.f90, `type manure_attributes` (variable `manure_om`)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one record per manure organic-matter profile.

#	Field	Type	Units	Description
1	name	char(64)	none	Profile name. Used as the crosswalk target from `manure_db.frt`.
2	frac_water	real	kg water / (kg manure + kg water)	Water fraction of the as-applied manure.
3	fcbn	real	kg C / kg manure	Carbon fraction.
4	fminn	real	kg min-N / kg manure	Mineral N fraction (NO3 + NH3).
5	fminp	real	kg min-P / kg manure	Mineral P fraction.
6	forgn	real	kg org-N / kg manure	Organic N fraction.
7	forgp	real	kg org-P / kg manure	Organic P fraction.
8	fnh3n	real	kg NH3-N / kg min-N	Share of mineral N that is NH3.
9	description	char(64)	none	Free-text description.

Example

`refdata/Osu_1hru/manure_om.frt`:

```
manure_om.frt: manure organic matter database
name            frac_water   fcbn          fminn         fminp         forgn         forgp         fnh3n   c
mw_beef_liq     0.988        0.000780576   0.000311751   0.00081295    0.002110312   0.000365707   0.99    r
mw_beef_slur    0.926        0.016786571   0.002697842   0.00081295    0.002110312   0.000365707   0.99    r
```

Two profiles. The liquid form is nearly all water (98.8%); the slurry form is 92.6% water with much higher solids and carbon.

Related files

- `chem_app.ops`. Fertilizer applications in management schedules reference `fertnm` from `fertilizer.frt` or a manure name from `manure_db.frt`.
- manure-allocation.md. Allocates manure from named source objects to HRUs and other receivers.
- codes.bsn. The active carbon model (`cswat`) controls how N from fertilizer and manure enters the soil pools.